

Lecture 16: Bonding in transition metal compounds and coordination complexes-1

Jin Xie (谢进), SPST, ShanghaiTech

Email: xiejin@shanghaitech.edu.cn

Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/12/12

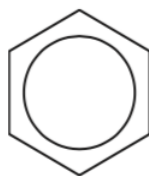
Major points of last lecture

- **Bonding in organic molecules**

1. Petroleum Refining and the Hydrocarbons
2. The Alkanes
3. The Alkenes and Alkynes
4. Aromatic Hydrocarbons
5. Fullerenes
6. Functional Groups and Organic Reactions

Aromatic Hydrocarbons 芳香烃

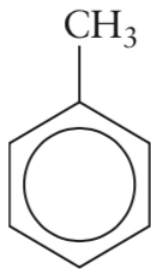
Aromatic compounds: **contain benzene rings** with various side groups or two (or more) benzene rings linked by alkyl chains or fused side by side, as in naphthalene ($C_{10}H_8$):



Benzene (苯)

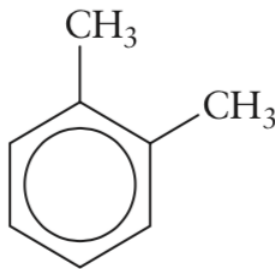


Naphthalene (萘)



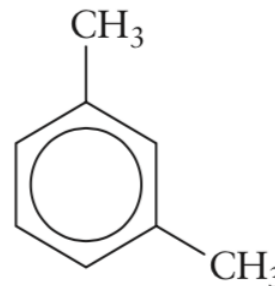
Toluene

甲苯



o-Xylene

邻二甲苯



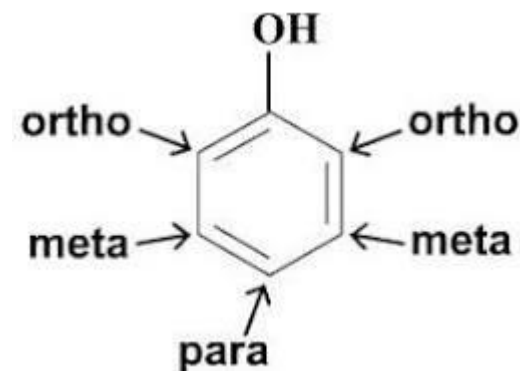
m-Xylene

间二甲苯



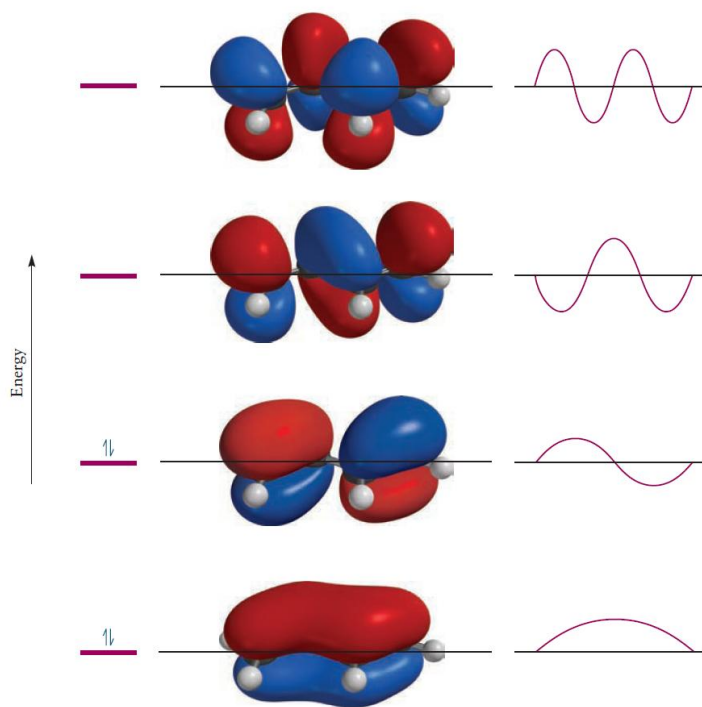
p-Xylene

对二甲苯

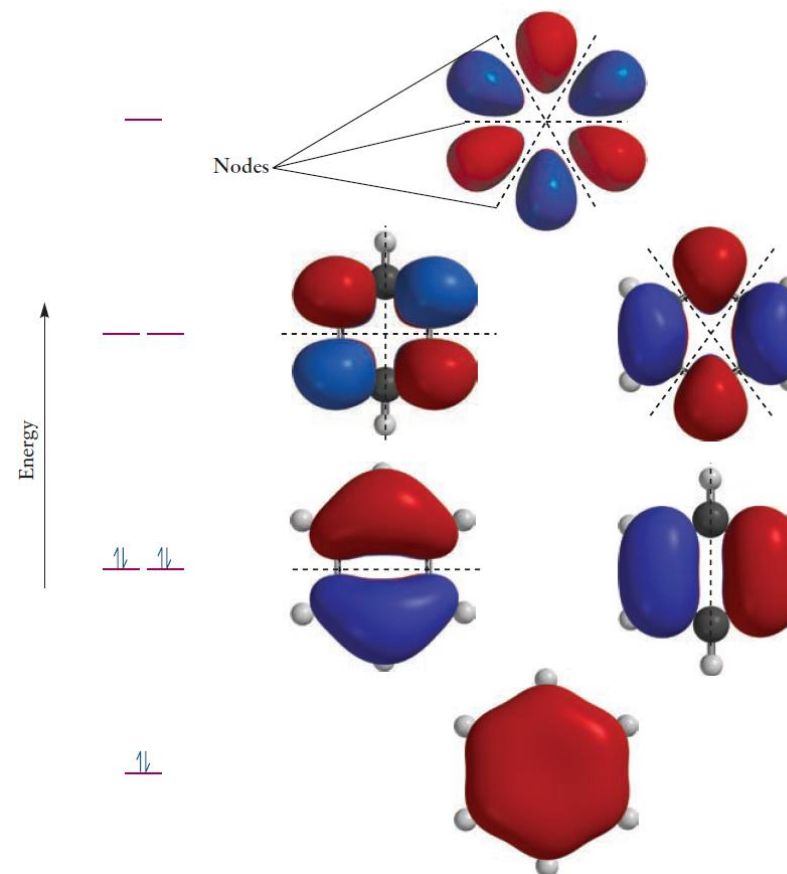


Aromatic Hydrocarbons

- ① Compatible orbital symmetries
- ② Maximum orbital overlap
- ③ Approximate atomic orbital energies



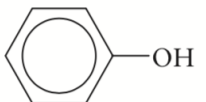
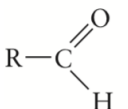
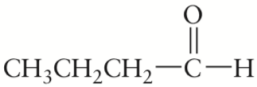
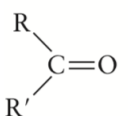
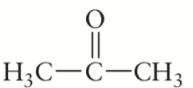
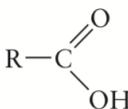
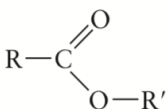
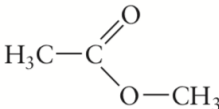
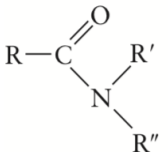
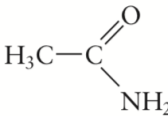
4 orbital, 4 electrons



6 orbital, 6 electrons

Functional Groups (官能团) and Organic Reactions

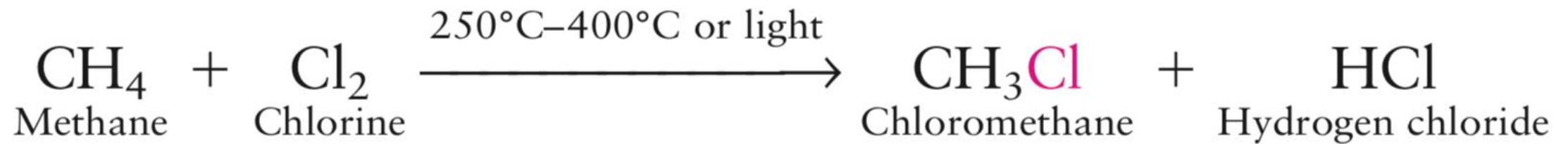
TABLE 7.3 Common Functional Groups

Functional Group†	Type of Compound	Examples
$R-F, -Cl, -Br, -I$	Alkyl or aryl halide	CH_3CH_2Br (bromoethane)
$R-OH$	Alcohol	CH_3CH_2OH (ethanol)
	Phenol	 (phenol)
$R-O-R'$	Ether	$H_3C-O-CH_3$ (dimethyl ether)
	Aldehyde	 (butyraldehyde, or butanal)
	Ketone	 (propanone, or acetone)
	Carboxylic acid	CH_3COOH (acetic acid, or ethanoic acid)
	Ester	 (methyl acetate)
$R-NH_2$	Amine	CH_3NH_2 (methylamine)
	Amide	 (acetamide)

Functional groups: the **reactive sites**, and their chemical properties depend only weakly on the natures of the hydrocarbons.

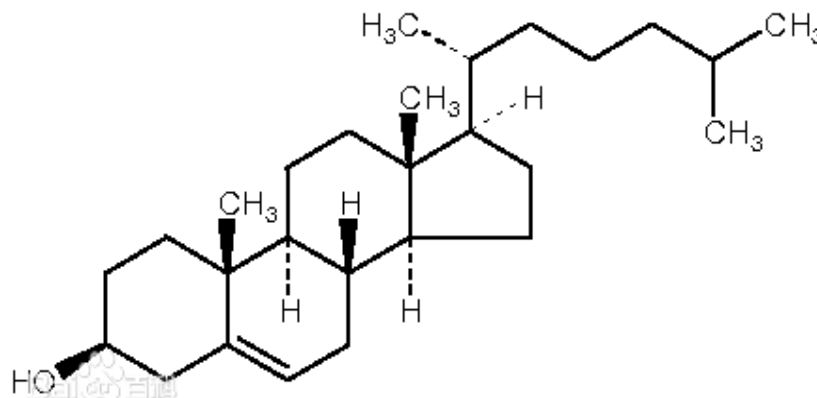
Halides (卤化物)

The Cl atom forms a σ bond to a C atom by overlap of its $3p_z$ orbital with a hybridized orbital on the carbon (sp^3 , sp^2 , or sp depending on the bonding in the hydrocarbon frame).

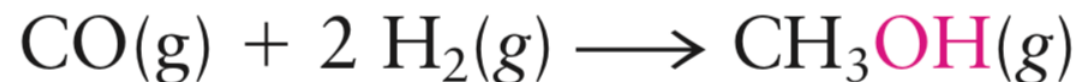


Alcohols醇类

Alcohols have the **-OH functional group attached to a tetrahedral carbon atom**

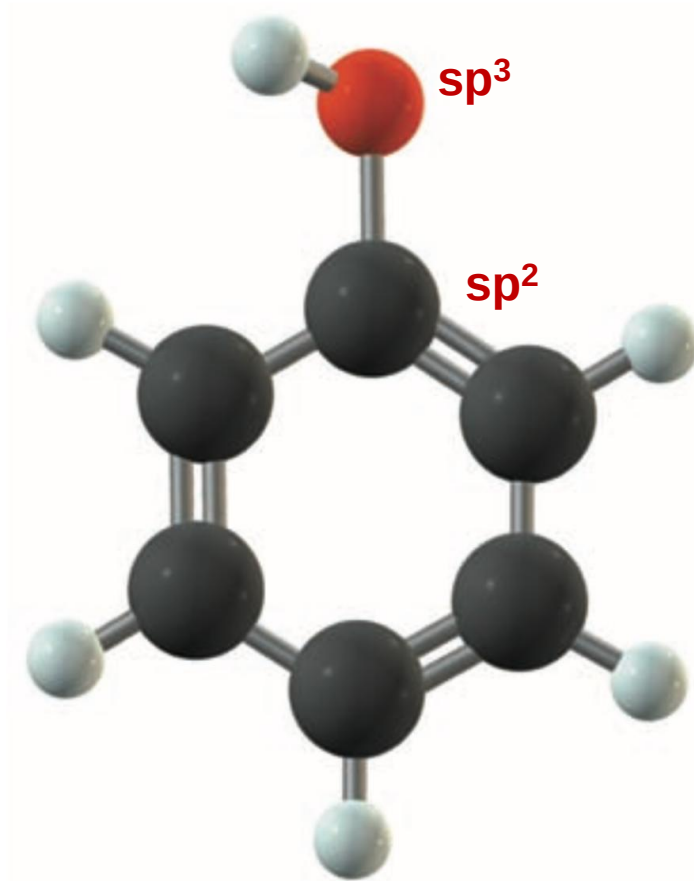


The simplest alcohol is methanol (CH₃OH), which is made from methane in a two-step process.

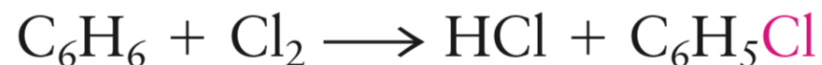


Phenols 苯酚类

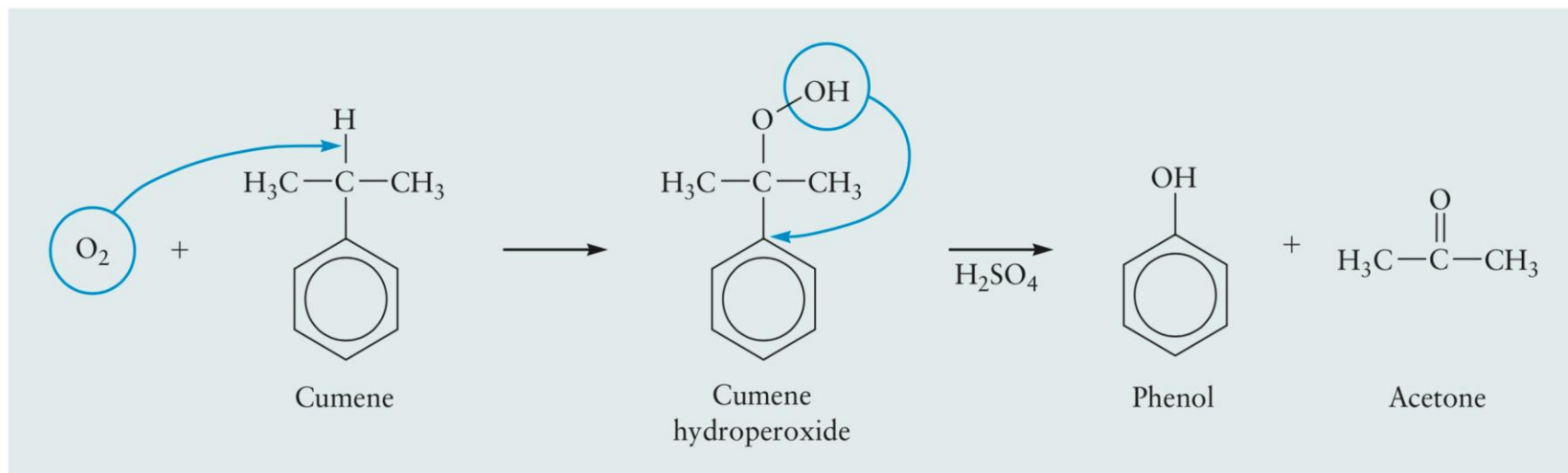
Phenols (苯酚类) are compounds in which an **OH group is attached directly to an aromatic ring**



The structure of phenol, C₆H₅OH.



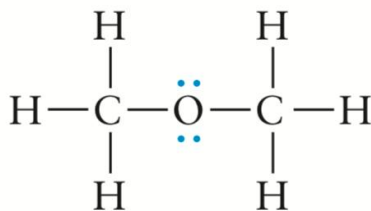
- When chlorine reacts with an alkene, it adds across the double bond.
- When an aromatic ring is involved, substitution of chlorine for hydrogen.



Ethers 醚类

Ethers are characterized by the **-O- functional group**, in which an oxygen atom provides a link between **two separate alkyl or aromatic groups**.

(a)



It can be produced by a condensation reaction.

(b)

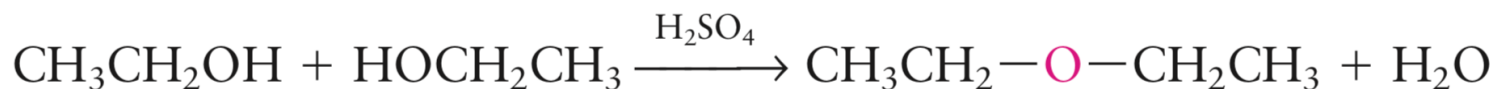
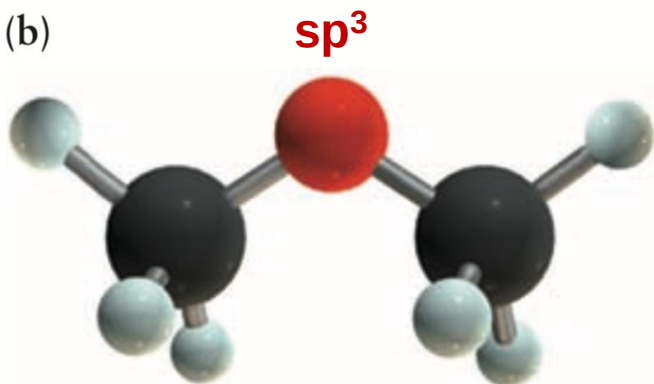
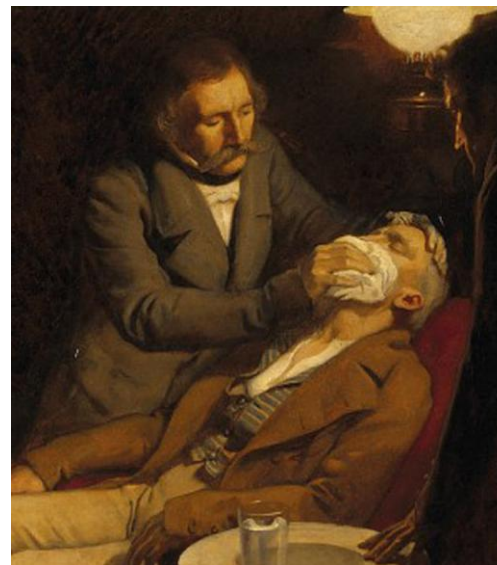


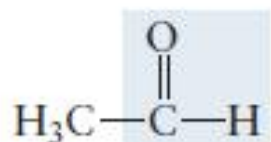
FIGURE 7.24 Structure of dimethyl ether. (a) Lewis diagram. (b) Ball-and-stick model.



Anesthesia 麻醉术

Aldehydes 醛类

An **aldehyde** contains the characteristic functional group $\text{—}\overset{\text{O}}{\parallel}\text{C—H}$ in its molecules



Acetaldehyde
(an aldehyde)

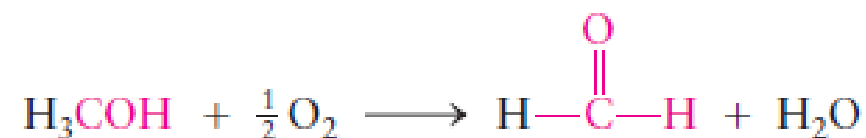
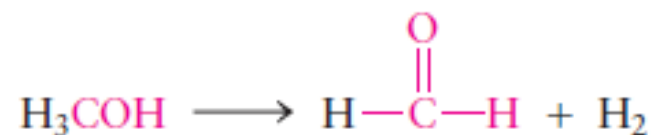
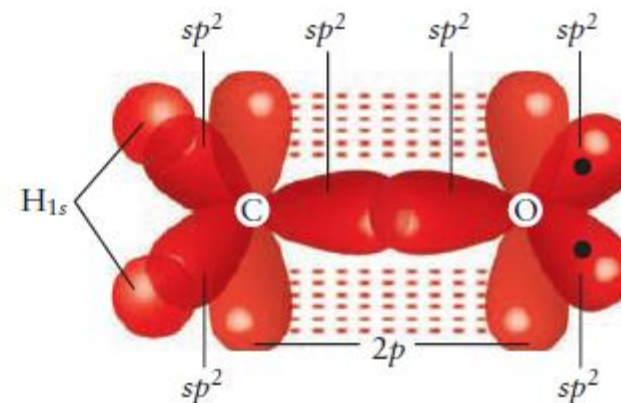


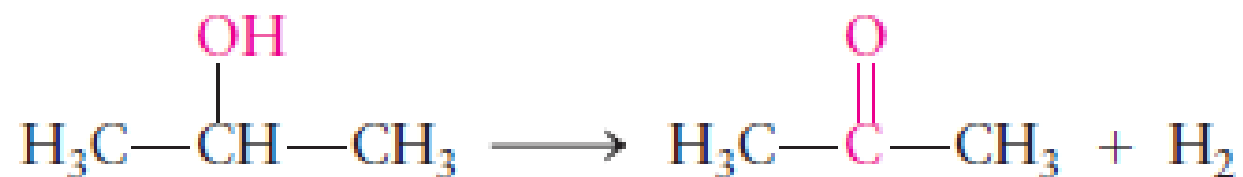
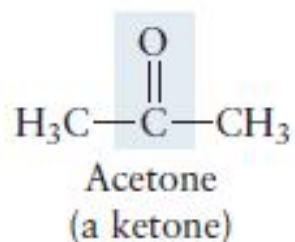
FIGURE 7.23 Bonding in formaldehyde involves sp^2 hybrid orbitals of both the carbon (C) and oxygen (O) atoms. The C—O σ bond is constructed by overlap of C and O sp^2 hybrid orbitals and the C—H bonds from overlap of C sp^2 orbitals with H 1s orbitals. Two lone pairs occupy the other O sp^2 hybrid orbitals. The π bond is formed by overlap of the parallel unhybridized 2p orbitals.



Both C and O are sp^2 hybridized.

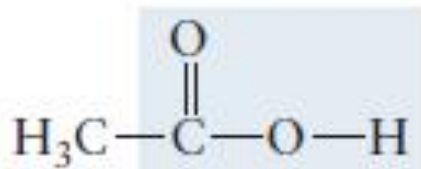
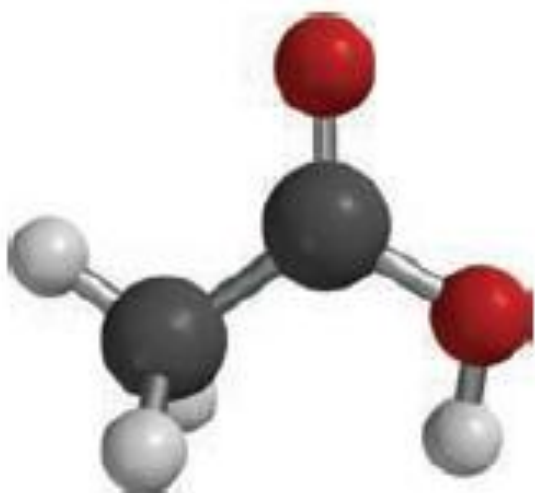
Ketones 酮类

Ketones have the $\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—} \end{array}$ functional group in which a carbon atom forms a double bond to an oxygen atom and single bonds to two separate alkyl or aromatic groups

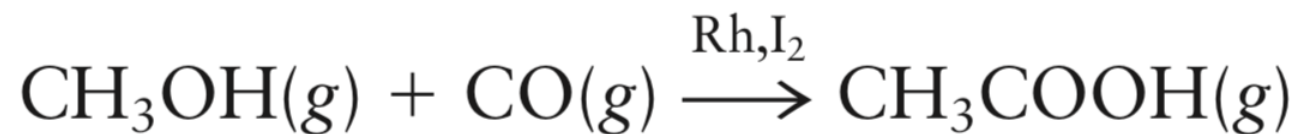
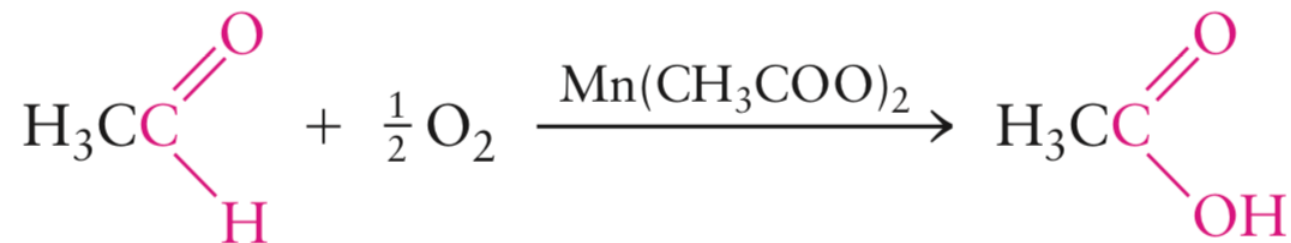


Carboxylic Acids 羧酸

Carboxylic acids contain the $\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—OH} \end{array}$ functional group (also written as -COOH).

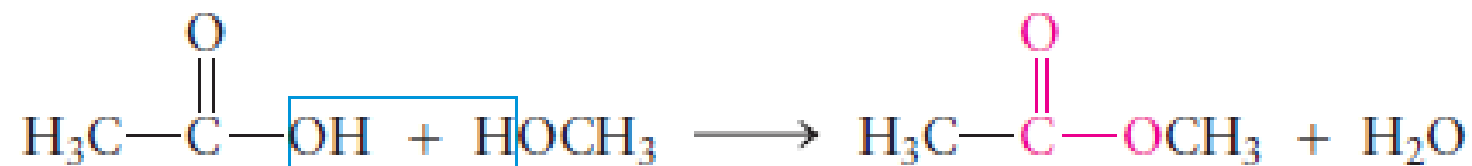
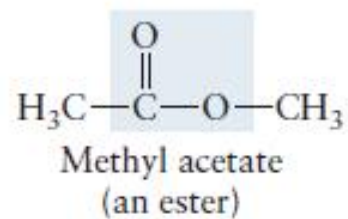
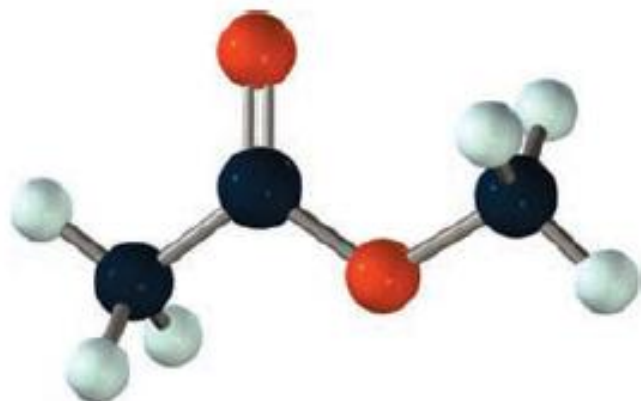


Acetic acid
(a carboxylic acid)



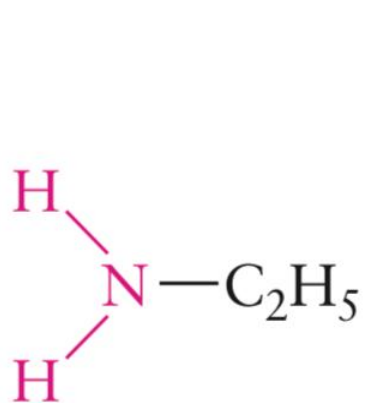
Esters 羧酸酯

Carboxylic acids react with alcohols or phenols to give **esters**, forming water as the coproduct



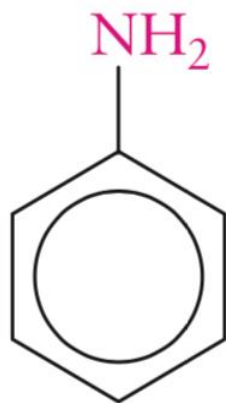
Amines 胺类

The **amines** are derivatives of ammonia with the general formula R_3N , where R can represent a hydrocarbon group or hydrogen.



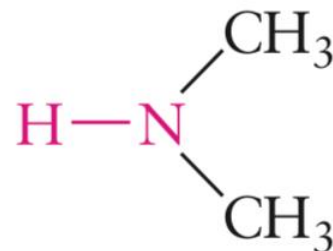
Ethylamine

primary amine



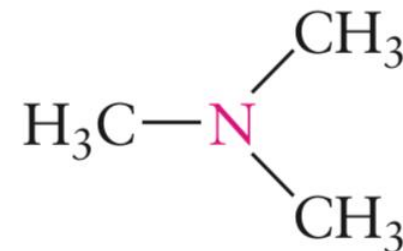
Aniline

primary amine



Dimethylamine

secondary amine



Trimethylamine

tertiary amine

N is sp^3 hybridized

Amides 酰胺

A primary or secondary amine (or ammonia itself) can react with a carboxylic acid to form an **amide**

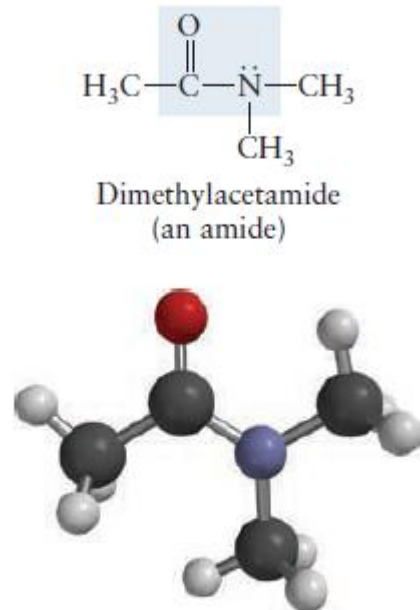
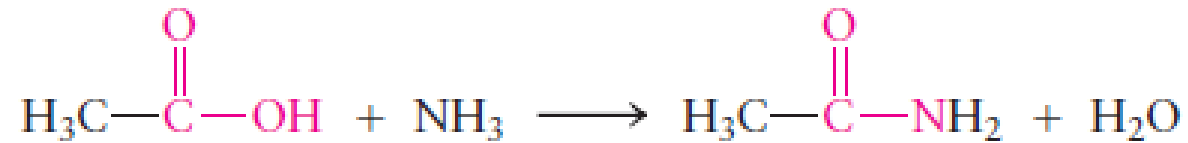
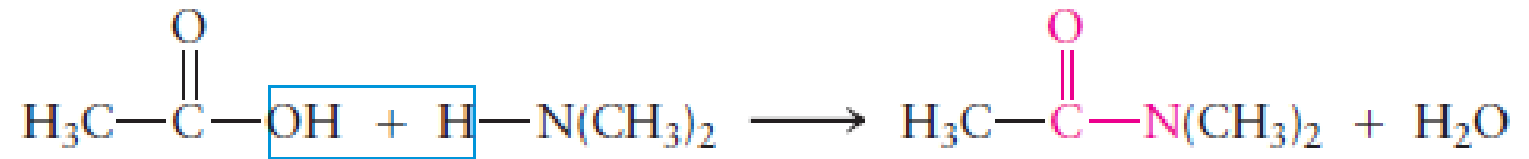


FIGURE 7.25 Bonding in dimethyl acetamide, an amide. The amide linkage is planar.



Amide linkages are present in the backbone of every protein molecule and are very important in biochemistry, where the structure of the molecule strongly influences its function

Lecture 16: Bonding in transition metal compounds and coordination complexes-1

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Major points of today's lecture

- **Bonding in transition metal compounds and coordination complexes**

1. **Chemistry of the Transition Metals**
2. **Introduction to Coordination Chemistry**
3. **Structures of Coordination Complexes**
4. **Crystal Field Theory: Optical and Magnetic Properties**
5. **Optical Properties and the Spectrochemical Series**
6. **Bonding in Coordination Complexes**

Chemistry of the Transition Metals (过渡金属)

过渡金属元素：元素周期表中d区（ds区包括周期表第IB ~ IIB族元素）。

（不包括镧系和锕系元素，周期表中57~71号元素叫做4f内过渡元素，89~103号元素叫做5f内过渡元素）

Transition metals in the d-block										
Group	3	4	5	6	7	8	9	10	11	12
Period 4	21Sc	22Ti	23V	24Cr	25Mn	26Fe	27Co	28Ni	29Cu	30Zn
5	39Y	40Zr	41Nb	42Mo	43Tc	44Ru	45Rh	46Pd	47Ag	48Cd
6	57La	72Hf	73Ta	74W	75Re	76Os	77Ir	78Pt	79Au	80Hg
7	89Ac	104Rf	105Db	106Sg	107Bh	108Hs	109Mt	110Ds	111Rg	112Cn

Chemistry of the Transition Metals

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	IA	IIA	IIIB	IVB	VB	VIB	VII	VIII	VIII	VIII	IB	IIB	IIIA	IVA	VA	VIA	VIIA	VIIIA
1	H																	He
2	Li	Be											B	C	N	O	F	Ne
3	Na	Mg											Al	Si	P	S	Cl	Ar
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
6	Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra	**															

Transition metals

* La Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu

** Ac Th Pa U Np Pu Am Cm Bk Cf Es Fm Md No Lr

The **partially filled d-electron shells** of the transition metal elements are responsible for a range of physical properties and chemical reactions **different from those of the main-group elements**.

Chemistry of the Transition Metals

First (3d) d-block Series (Sc–Zn)

Group	3	4	5	6	7	8	9	10	11	12
Atomic nr	21	22	23	24	25	26	27	28	29	30
Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Electron configuration	$3d^1 4s^2$	$3d^2 4s^2$	$3d^3 4s^2$	$3d^5 4s^1$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^8 4s^2$	$3d^{10} 4s^1$	$3d^{10} 4s^2$

The energies of the 4s and 3d orbitals are so close: the electron configurations of the neutral atoms and their ions are not easily predicted

Second (4d) d-block Series (Y–Cd)

Atomic nr	39	40	41	42	43	44	45	46	47	48
Element	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
Electron configuration	$4d^1 5s^2$	$4d^2 5s^2$	$4d^4 5s^1$	$4d^5 5s^1$	$4d^5 5s^2$	$4d^7 5s^1$	$4d^8 5s^1$	$4d^{10}$	$4d^{10} 5s^1$	$4d^{10} 5s^2$

Chemistry of the Transition Metals

Third (5d) d-block Series (La–Hg)

Atomic nr	57	72	73	74	75	76	77	78	79	80
Element	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
Electron configuration	$5d^1 6s^2$	$5d^2 6s^2$	$5d^3 6s^2$	$5d^4 6s^2$	$5d^5 6s^2$	$5d^6 6s^2$	$5d^7 6s^2$	$5d^9 6s^1$	$5d^{10} 6s^1$	$5d^{10} 6s^2$

Fourth (6d) d-block Series (Ac–Cn)

Atomic nr	89	104	105	106	107	108	109	110	111	112
Element	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn
Electron configuration	$6d^1 7s^2$	$6d^2 7s^2$	$6d^3 7s^2$	$6d^4 7s^2$	$6d^5 7s^2$	$6d^6 7s^2$	$6d^7 7s^2$	$6d^8 7s^2$	$6d^9 7s^2$	$6d^{10} 7s^2$

Chemistry of the Transition Metals: physical properties

Increasing Z_{eff} from left to right along each period: increasing electron–nuclear attraction

T A B L E 8.1

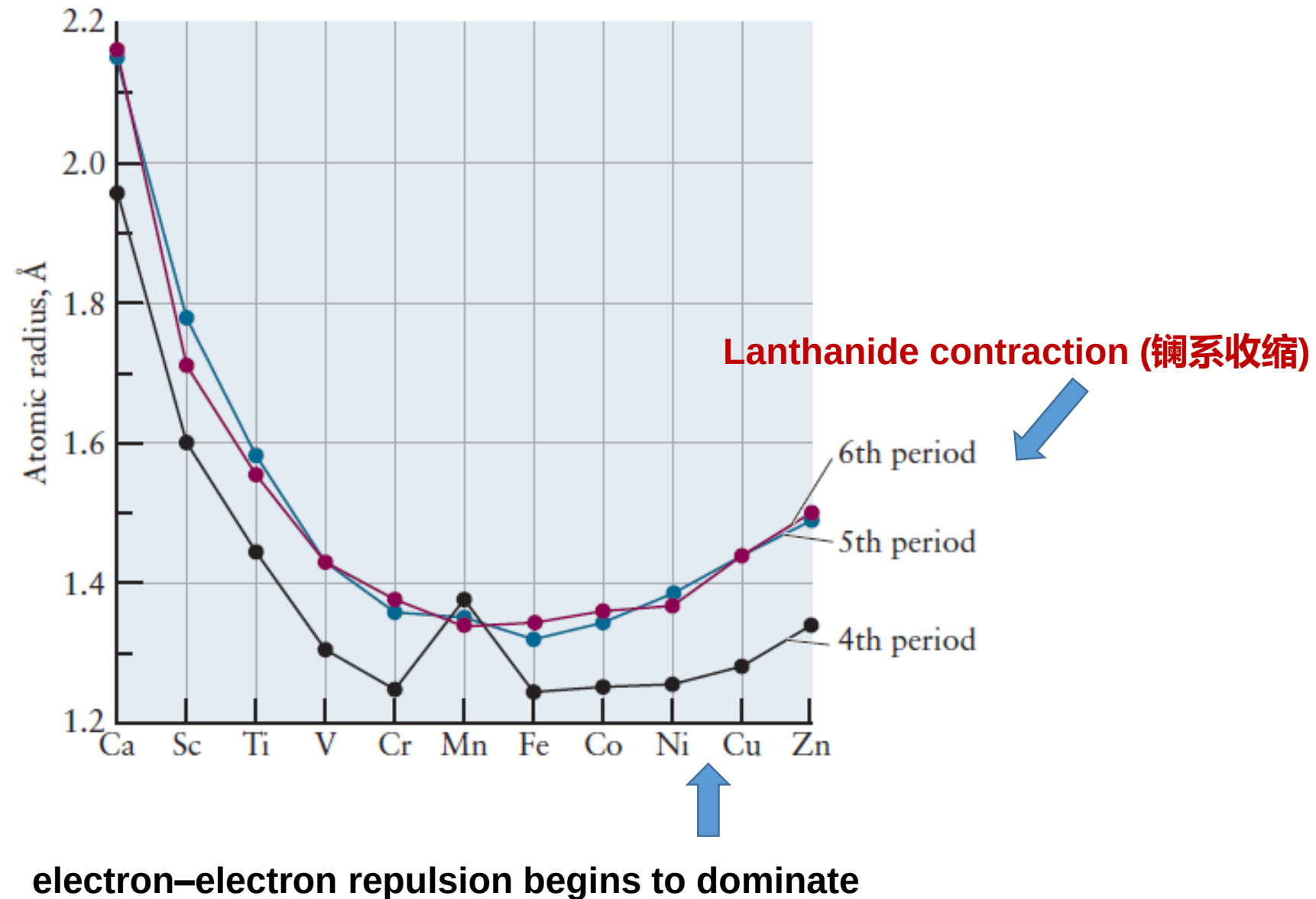
Properties of the Fourth-Period Transition Elements

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
IE_1 (kJ mol ^{−1})	631	658	650	653	717	759	758	737	745	906
IE_2 (kJ mol ^{−1})	1235	1310	1414	1592	1509	1562	1648	1753	1958	1733
Boiling point (°C)	2831	3287	3380	2672	1962	2750	2870	2732	2567	907
Melting point (°C)	1541	1660	1890	1857	1244	1535	1495	1453	1083	420
Atomic radius (Å)	1.61	1.45	1.31	1.25	1.37	1.24	1.25	1.25	1.28	1.34
M ²⁺ ionic radius (Å)	0.81	0.68	0.88	0.89	0.80	0.72	0.72	0.69	0.72	0.74
M ²⁺ configuration	d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{10}
$\Delta H_{\text{hyd}}(\text{M}^{2+})^\dagger$ (kJ mol ^{−1})				−2799	−2740	−2839	−2902	−2985	−2989	−2937

[†]Defined as $\Delta H_f^\circ(\text{M}^{2+}(\text{aq}) - \Delta H_f^\circ(\text{M}^{2+}(\text{g}))$.

Chemistry of the Transition Metals: physical properties

FIGURE 8.1 Variation of atomic radii through the fourth-, fifth-, and sixth-period transition-metal elements. The chemical symbols shown at the bottom of the graph refer to the fourth-period elements.



Chemistry of the Transition Metals: physical properties

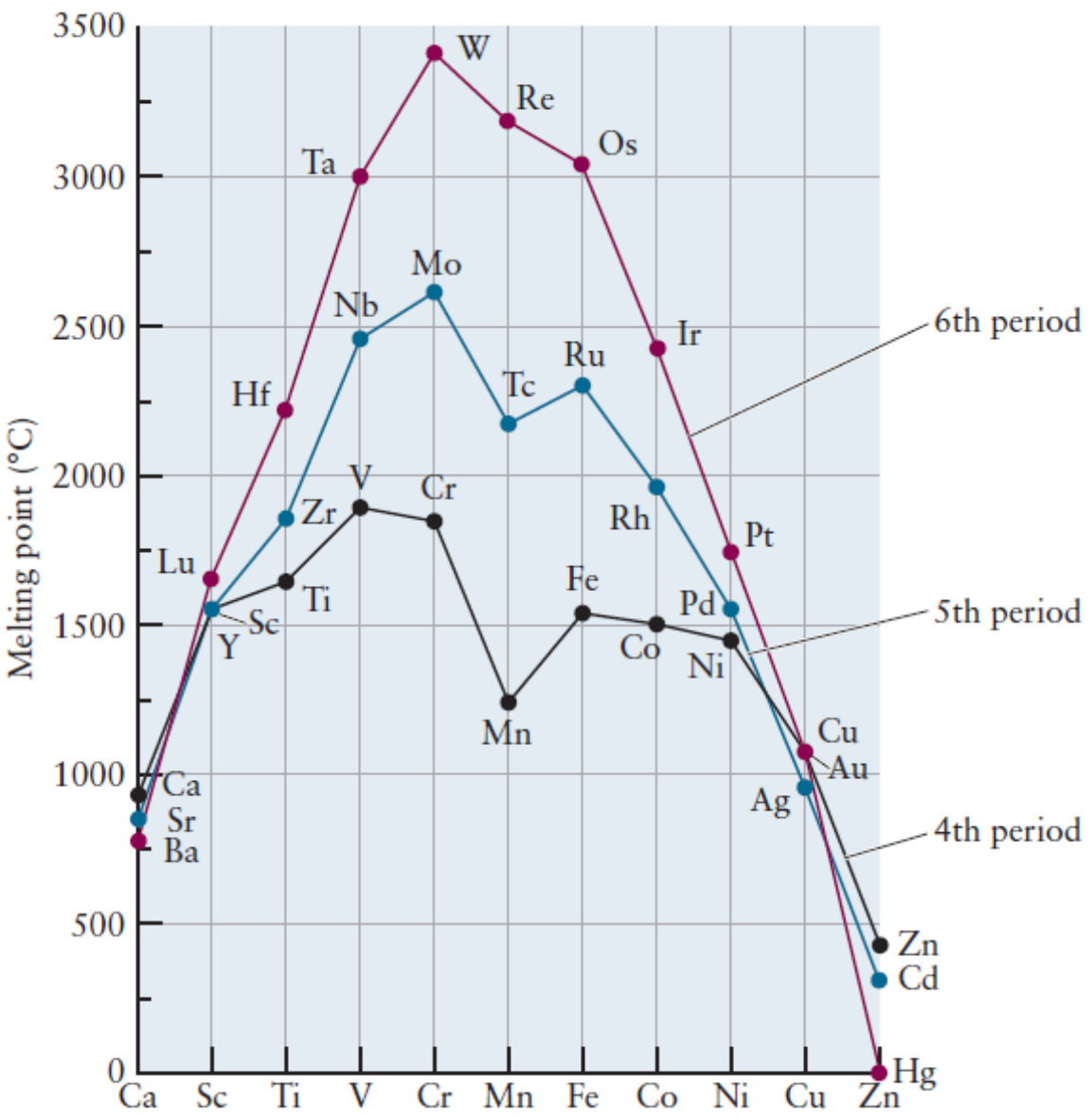
H 25																	He
Li 145	Be 105	Atomic radius: pm										B 85	C 70	N 65	O 60	F 50	Ne
Na 180	Mg 150											Al 125	Si 110	P 100	S 100	Cl 100	Ar
K 220	Ca 180	Sc 160	Ti 140	V 135	Cr 140	Mn 140	Fe 140	Co 135	Ni 135	Cu 135	Zn 135	Ga 130	Ge 125	As 115	Se 115	Br 115	Kr
Rb 235	Sr 200	Y 180	Zr 155	Nb 145	Mo 145	Tc 135	Ru 130	Rh 135	Pd 140	Ag 160	Cd 155	In 155	Sn 145	Sb 145	Te 140	I 140	Xe
Cs 260	Ba 215	*	Hf 155	Ta 145	W 135	Re 135	Os 130	Ir 135	Pt 135	Au 135	Hg 150	Tl 190	Pb 180	Bi 160	Po 190	At	Rn
Fr	Ra 215	**	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og
*	La 195	Ce 185	Pr 185	Nd 185	Pm 185	Sm 185	Eu 185	Gd 180	Tb 175	Dy 175	Ho 175	Er 175	Tm 175	Yb 175	Lu 175		
**	Ac 195	Th 180	Pa 180	U 175	Np 175	Pu 175	Am 175	Cm	Bk	Cf	Es	Fm	Md	No	Lr		

Atomic radius: pm

Lanthanide contraction:

- the first and second transition series are separated by 18 elements, whereas the second and third are separated by 32 elements.
- but the f orbitals are much more diffuse than the d orbitals and much less effective in screening the nuclear charge.

Chemistry of the Transition Metals: physical properties

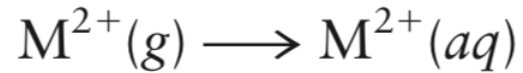


金属键 (metallic bond) 是化学键的一种，
主要在金属中存在。

Metal-metal bond strengths **first increase and then decrease going across each period**, reaching a maximum in the middle.

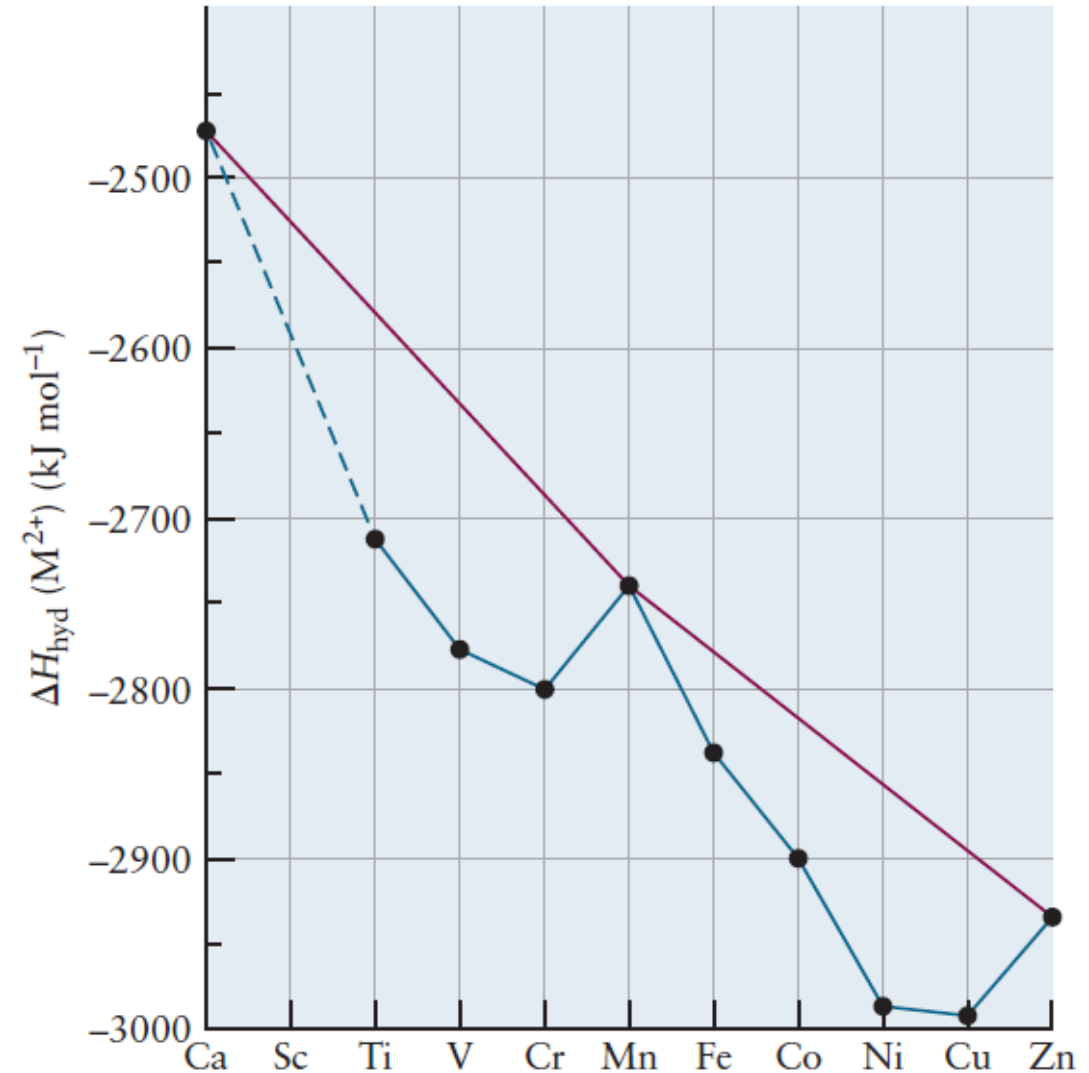
(the number of unpaired electrons in the elements, which reaches a maximum in the middle of the corresponding series.)

Chemistry of the Transition Metals: physical properties



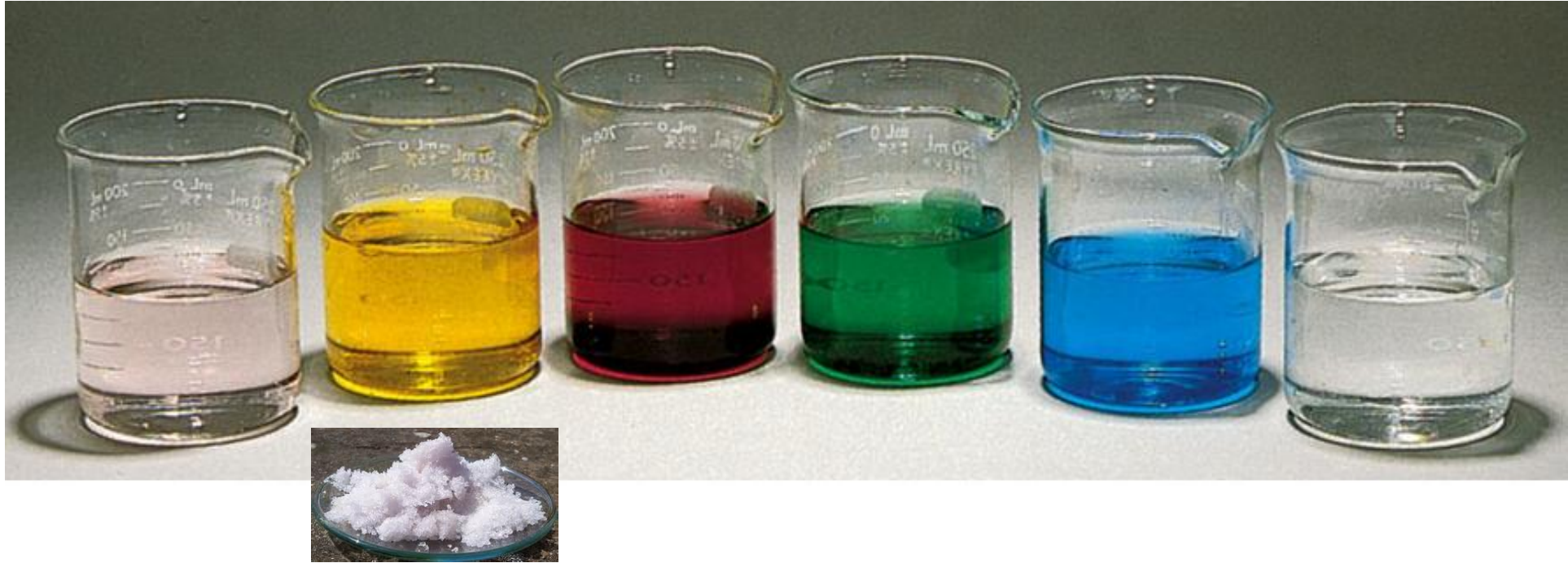
$[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ ions $[\text{Ni}(\text{NH}_3)_6]^{2+}$ ions

A metal ion is **hydrated (水和)**: the energy change is related to changes in the thermodynamic function **enthalpy (焓)**, denoted by **H**.



strength of the interactions to **increase as the ionic radii decrease**, allowing the water molecules to approach the ions more closely.

Hydrated Transition Metal Ions



$\text{Mn}(\text{H}_2\text{O})_6^{2+}$	$\text{Fe}(\text{H}_2\text{O})_6^{3+}$	$\text{Co}(\text{H}_2\text{O})_6^{2+}$	$\text{Ni}(\text{H}_2\text{O})_6^{2+}$	$\text{Cu}(\text{H}_2\text{O})_4^{2+}$	$\text{Zn}(\text{H}_2\text{O})_4^{2+}$
5	5	7	8	9	10
12	12	12	12	8	8
17	17	19	20	17	18

Chemistry of the Transition Metals: Oxidation States

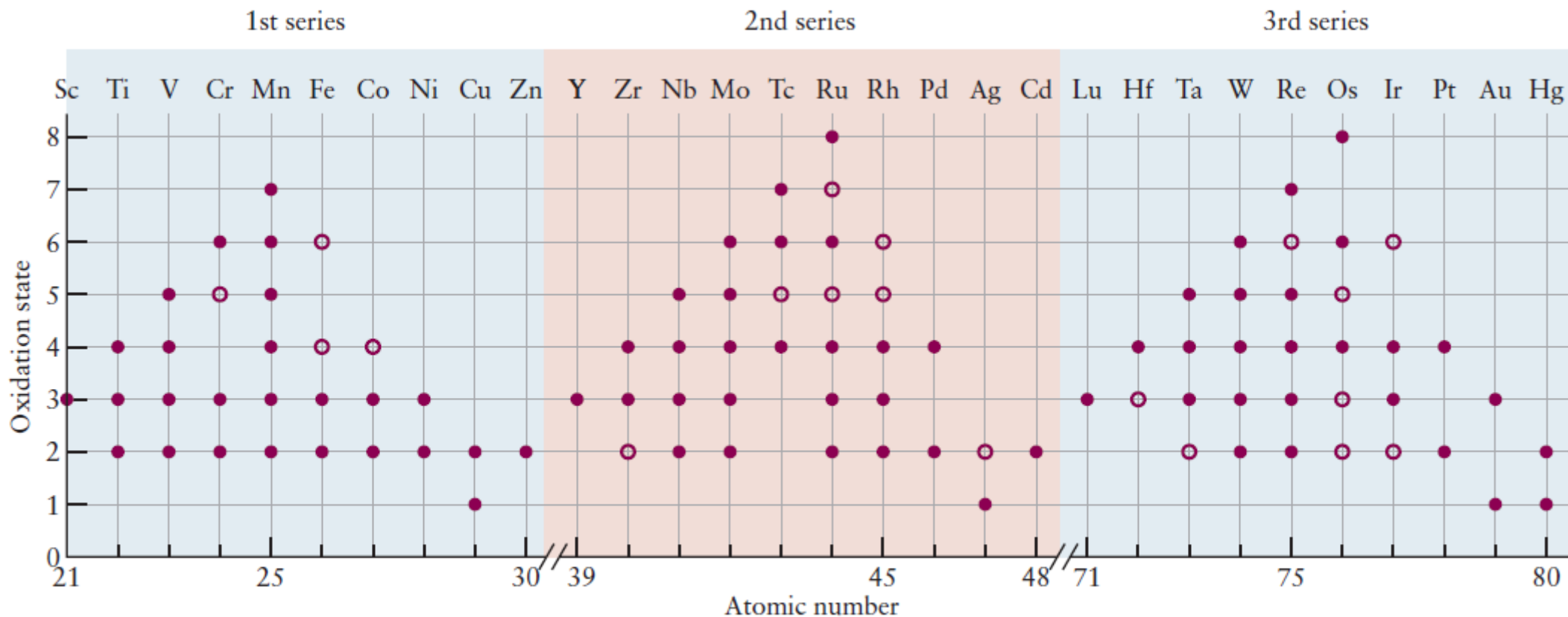


FIGURE 8.4 Some of the oxidation states found in compounds of the transition-metal elements. The more common oxidation states are represented by solid circles, and the less common ones are represented by open circles.

The maximum oxidation states of the **early members** of each period result from the participation of **all the outer s and d electrons in ionic or covalent bonding**.

Chemistry of the Transition Metals: Oxidation States

Compounds with transition metals:

in **high oxidation** states: relatively **covalent**;

in **low oxidation** states: relatively **ionic**.

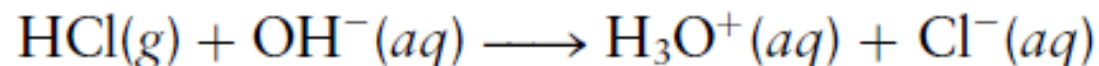
- Mn_2O_7 : covalent compound; it is a liquid at room temperature (crystallizing only at 6°C);
- Mn_3O_4 : ionic compound; containing both Mn(II) and Mn(III), melts at 1564°C .
- A wide range of oxidations states: **partially filled d orbitals** can either **accept or donate electrons** to form chemical bonds.
- Many of their compounds are effective catalysts (催化剂), substances that increase the rates of chemical reactions without undergoing permanent chemical changes themselves.

Chemistry of the Transition Metals: Hard and soft acids and bases

Brønsted-Lowry definition:

Acids (酸): proton donors;

Bases (碱): proton acceptors



Lewis concept:

Acids: electron acceptors;

Bases: electron donors



Hard acid (硬酸): High charge density: small, multiply charged ion

Soft acid (软酸): Low charge density: large, low oxidation state

Chemistry of the Transition Metals: Hard and soft acids and bases

T A B L E 8.2

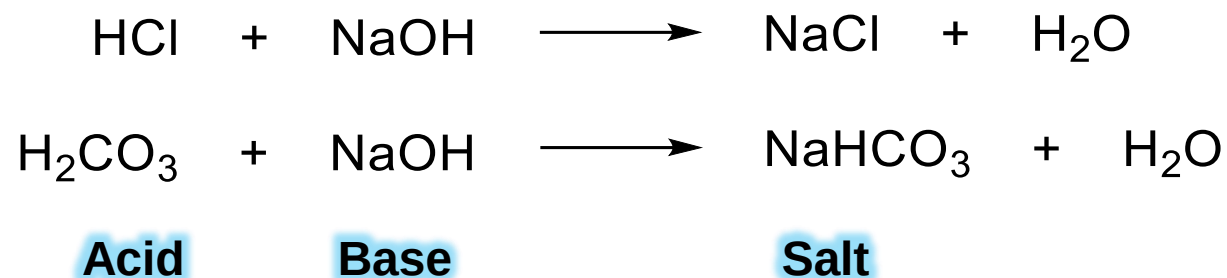
Classification of Lewis Acids and Bases[†]

	Hard	Borderline	Soft
Acids	$\text{H}^+, \text{Li}^+, \text{Na}^+, \text{K}^+$ $\text{Be}^{2+}, \text{Mg}^{2+}, \text{Ca}^{2+}$ $\text{Cr}^{2+}, \text{Cr}^{3+}, \text{Al}^{3+}$ SO_3, BF_3	$\text{Fe}^{2+}, \text{Co}^{2+}, \text{Ni}^{2+}$ $\text{Cu}^{2+}, \text{Zn}^{2+}, \text{Pb}^{2+}$ $\text{SO}_2, \text{BBr}_3$	$\text{Cu}^+, \text{Ag}^+, \text{Au}^+, \text{Tl}^+, \text{Hg}^+$ $\text{Pd}^{2+}, \text{Cd}^{2+}, \text{Pt}^{2+}, \text{HG}^{2+}$ BH_3
Bases	$\text{F}^-, \text{OH}^-, \text{H}_2\text{O}, \text{NH}_3$ $\text{CO}_3^{2-}, \text{NO}_3^-, \text{O}^{2-}$ $\text{SO}_4^{2-}, \text{PO}_4^{3-}, \text{ClO}_4^-$	$\underline{\text{NO}}_2^-, \underline{\text{SO}}_3^{2-}, \text{Br}^-$ N_3^-, N_2 $\text{C}_6\text{H}_5\text{N}, \underline{\text{SCN}}^-$	$\text{H}^-, \text{R}^-, \underline{\text{CN}}^-, \underline{\text{CO}}, \text{I}^-$ $\text{CN}^-, \text{R}_3\text{P}, \text{C}_6\text{H}_6$ R_2S

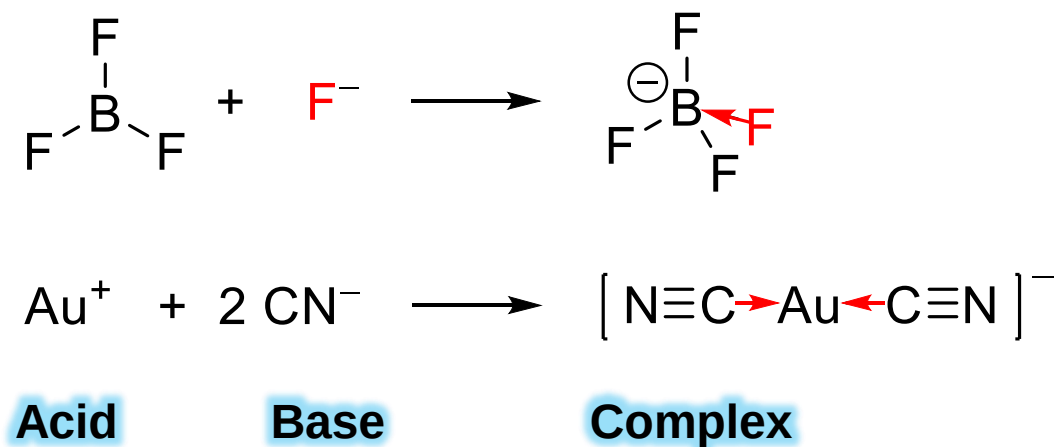
[†]The underlined element identifies the electron pair donor if there is more than one possible choice.

Hard and Soft Acids and Bases

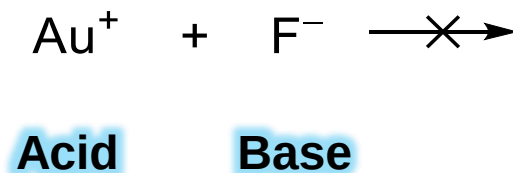
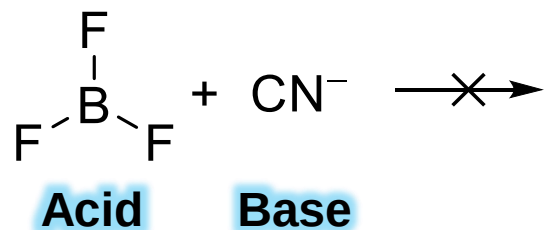
Protonic
acid / base



Lewis
acid / base



Hard and Soft Acids and Bases

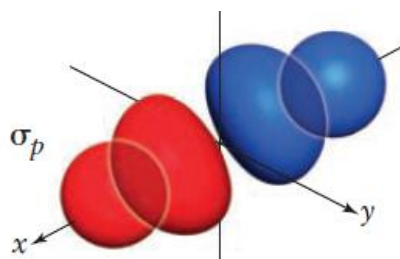


Hard: Electron **tightly** bound

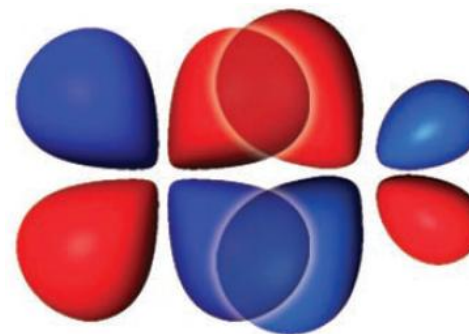
Soft: Electron **loosely** bound



VS



VS



HSAB

[illegible]

		
hard	intermediate	soft
C	N	O
	P	S
	As	Se
	Sb	Te
		I
		Br
		Cl

“硬亲硬,软亲软,软硬交界都不管”

Solubility: $\text{AgF} \gg \text{AgCl} > \text{AgBr} > \text{AgI} \approx \text{Ag}(\text{CN})_2 > \text{Ag}_2\text{S}$

$$\text{CsF} > \text{NaF} \gg \text{LiF} > \text{CaF}_2$$


Ralph G. Pearson
(1919–)

Assign oxidation states to the transition-metal ions in the following compounds: TiO_2 , ZrCl_4 , VCl_3 , CrF_6 , Mo_2S_3 , FeO , AuCl_3 . Explain, where appropriate, which ligands stabilize the higher oxidation states and why.

Use the hard/soft acid/base concept to predict whether the following reactions will occur.



Major points of today's lecture

- **Bonding in transition metal compounds and coordination complexes**
 1. Chemistry of the Transition Metals
 2. **Introduction to Coordination Chemistry**
 3. Structures of Coordination Complexes
 4. Crystal Field Theory: Optical and Magnetic Properties
 5. Optical Properties and the Spectrochemical Series
 6. Bonding in Coordination Complexes

Introduction to Coordination (配位) Chemistry

The Alsatian-Swiss chemist Alfred Werner pioneered the field of coordination chemistry in the late nineteenth century.

Compound 1: $\text{CoCl}_3 \cdot 6\text{NH}_3$ Orange-yellow

Compound 2: $\text{CoCl}_3 \cdot 5\text{NH}_3$ Purple

Compound 3: $\text{CoCl}_3 \cdot 4\text{NH}_3$ Green

Compound 4: $\text{CoCl}_3 \cdot 3\text{NH}_3$ Green

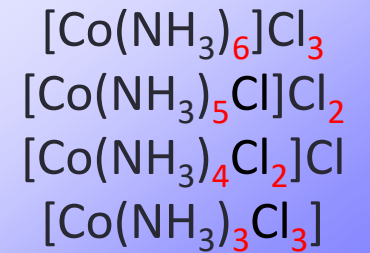


Alfred Werner
(1866–1919)

Aqueous hydrochloric acid did not remove the ammonia: ammonia was closely bound

Introduction to Coordination Chemistry

Compound 1: $\text{CoCl}_3 \cdot 6\text{NH}_3$	Orange-yellow	$+ \text{AgNO}_3 \longrightarrow \text{AgCl}$	100%
Compound 2: $\text{CoCl}_3 \cdot 5\text{NH}_3$	Purple		66.7%
Compound 3: $\text{CoCl}_3 \cdot 4\text{NH}_3$	Green		33.3%
Compound 4: $\text{CoCl}_3 \cdot 3\text{NH}_3$	Green		0%



Werner: **two different kinds of species** associated with the cobalt ions

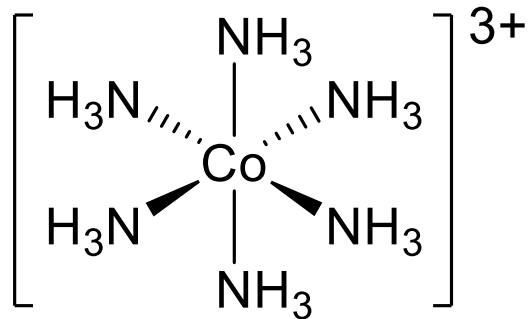
The **primary valences** (主价) were anions like Cl^- in simple salts such as CoCl_3 .

The primary valences were nondirectional. (outer coordination sphere 外界)

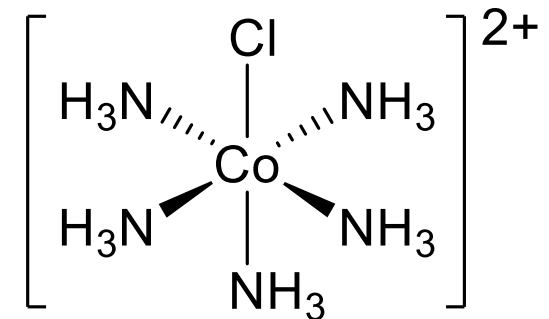
The **secondary valences** (副价) could be either simple anions or neutral molecules such as NH_3 .

The secondary valences were oriented along well-defined directions. (inner coordination sphere 内界)

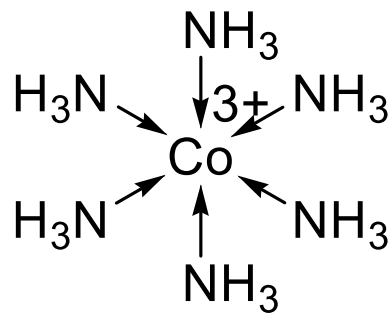
Cobaltammines 钴氨化合物



Hexaminecobalt(III)
六氨合钴(III)

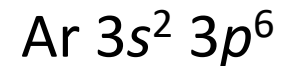
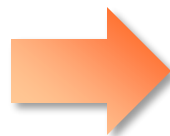
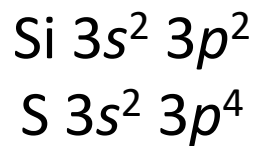
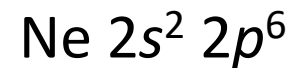
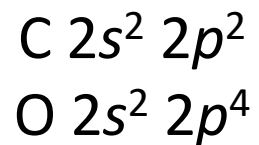


Pentamminechlorocobalt(III)
一氯五氨合钴(III)

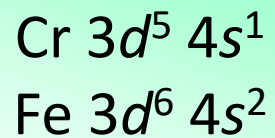


$$9 - 3 + 2 \times 6 = 18$$

The 18-Electron Rule (1921)

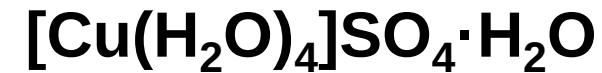
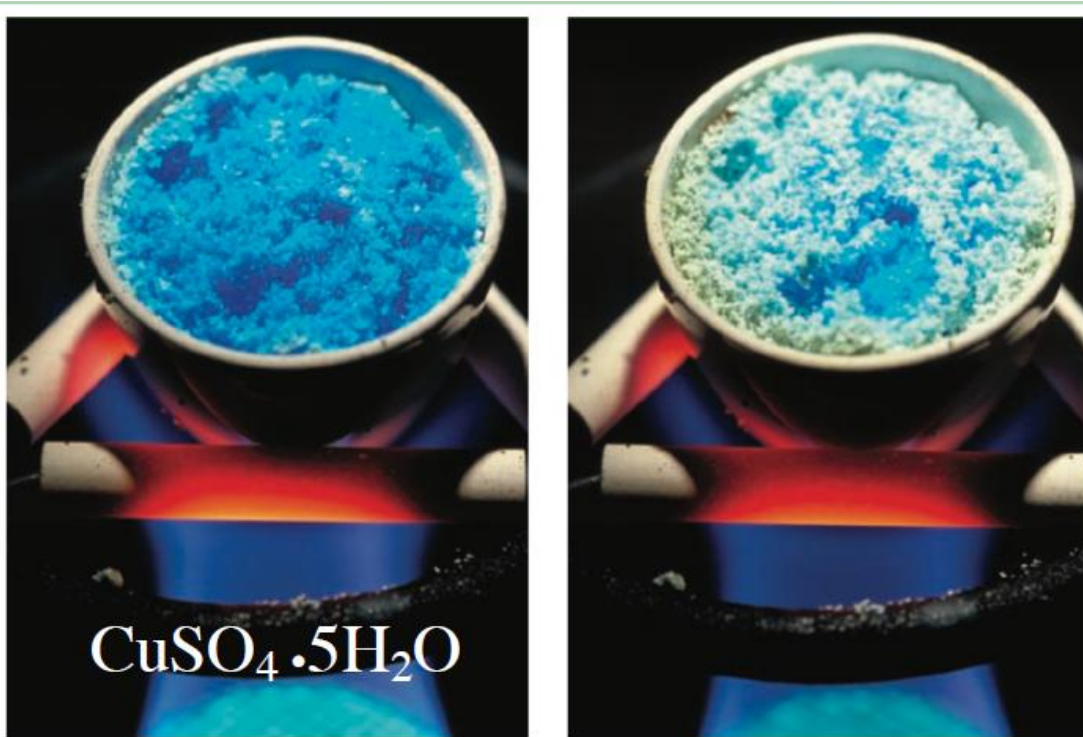


Octet rule



18-electron rule

Introduction to Coordination Chemistry



- The ability to make fairly strong, directional bonds by accepting electron pairs from neighboring molecules or ions is characteristic of the **transition-metal elements**.
- The solvation of Na^+ by H_2O molecules in aqueous solution is an example of weak coordination.

Introduction to Coordination Chemistry

Coordination modifies the chemical and physical properties of both the **central atom** and **the ligands**.

CN⁻ ion reacts immediately with acid to generate gaseous HCN, a deadly poison.



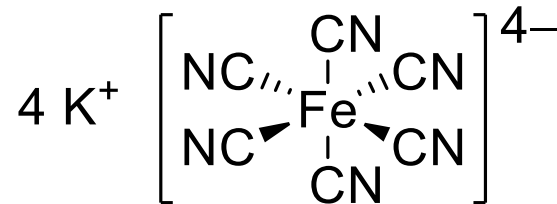
KCN



Fe²⁺, when mixed with aqueous base, instantly precipitates a gelatinous hydroxide.

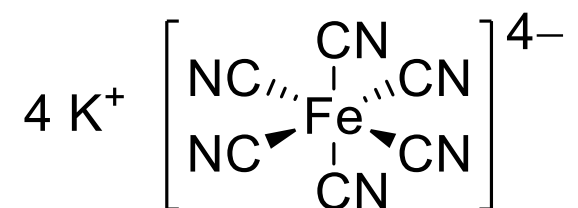


FeSO₄
· 7H₂O

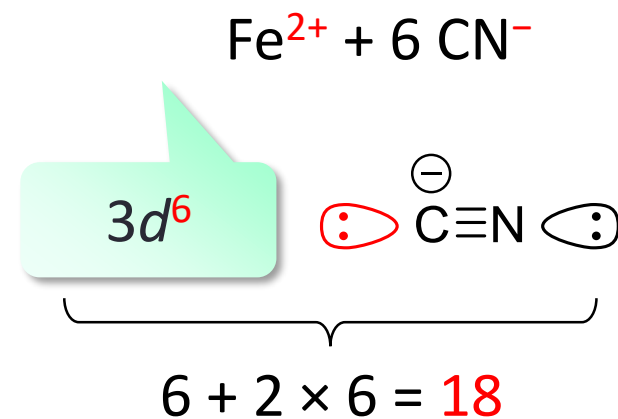
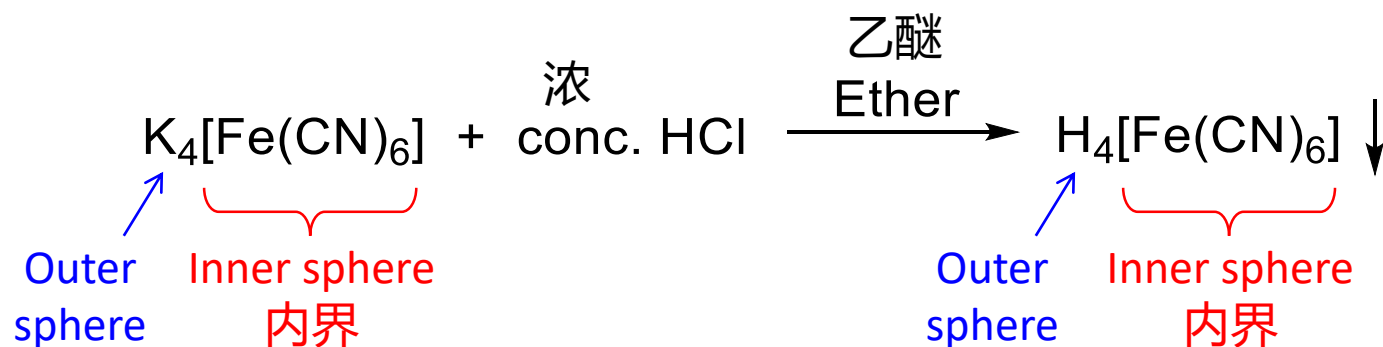


The reaction between Fe²⁺ and CN⁻ produces the complex ion [Fe(CN)₆]⁴⁻(aq), which undergoes neither of the two reactions just described.

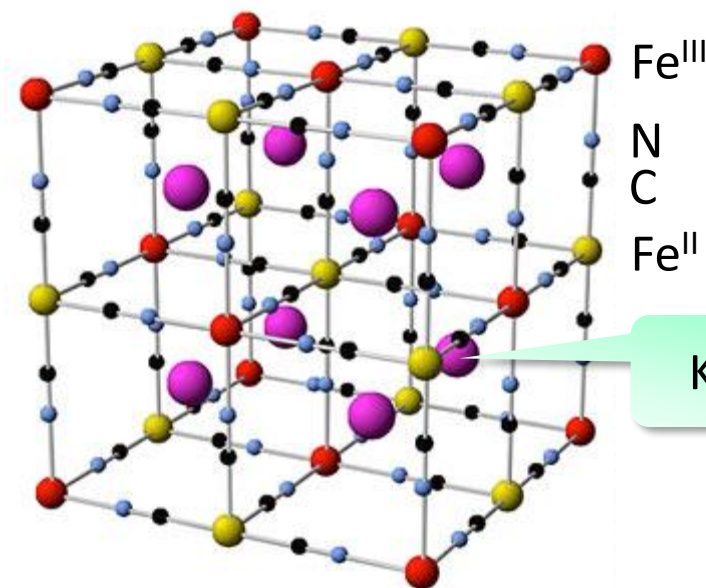
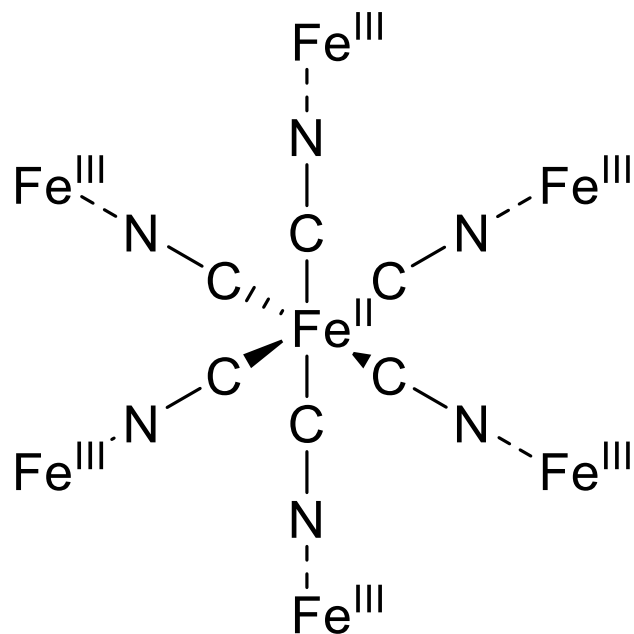
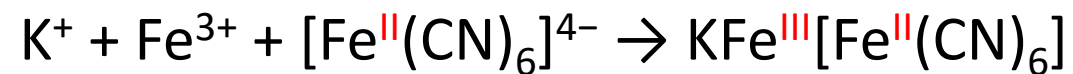
Ferrocyanide



$\text{K}_4[\text{Fe}(\text{CN})_6]$
 Potassium **ferro**cyanide
亚铁氰化钾, 黄血盐
 Potassium hexacyano**ferrate**(II)
 六氰合铁(II)酸钾

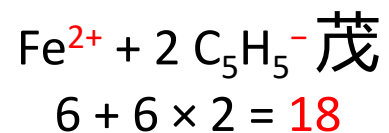
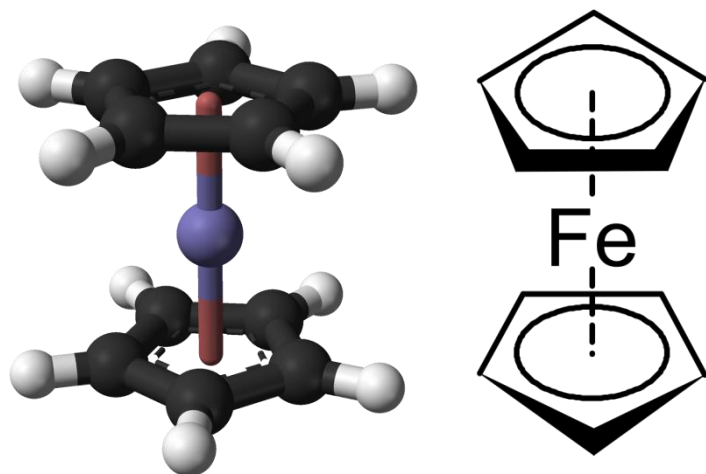


Prussian Blue 普鲁士蓝



K⁺, TI⁺, ...

Ferrocene 二茂铁



Geoffrey Wilkinson
(1921–1996)



Ernst Otto Fischer
(1918–2007)

Introduction to Coordination Chemistry

TABLE 8.3

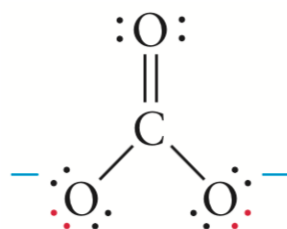
Common Ligands and Their Names

Ligand [†]	Name
:NO ₂ ⁻	Nitro
:OCO ₂ ²⁻	Carbonato
:ONO ⁻	Nitrito
:CN ⁻	Cyano
:SCN ⁻	Thiocyanato
:NCS ⁻	Isothiocyanato
:OH ⁻	Hydroxo
:OH ₂	Aqua
:NH ₃	Ammine
:CO	Carbonyl
:NO ⁺	Nitrosyl

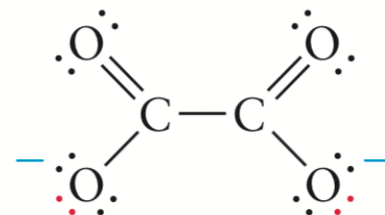
[†]The ligating atom is indicated by a pair of dots (:) to show a lone pair of electrons. In the CO₃²⁻ ligand, either one or both of the O atoms can donate a lone pair to a metal.

Ligands that bond to a metal atom through a single point of attachment are called **monodentate** (单齿) (derived from Latin *mono*, meaning “one,” plus *dens*, meaning “tooth,” indicating that they bind at only one point) **bidentate**, **tridentate**, and so forth.

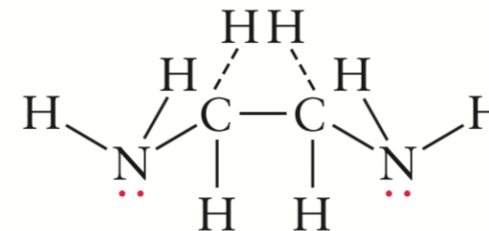
Complexes in which a ligand coordinates via two or more donors to the same central atom are called **chelates** (螯合物)



Carbonate ion,
CO₃²⁻



Oxalate ion,
C₂O₄²⁻



Ethylenediamine,
NH₂CH₂CH₂NH₂

Determine the oxidation state of the coordinated metal atom in each of the following compounds:

(a) $\text{K}[\text{Co}(\text{NH}_3)_2(\text{CN})_4]$; (b) $\text{Os}(\text{CO})_5$; (c) $\text{Na}[\text{Co}(\text{OH}_2)_3(\text{OH})_3]$.

Naming Coordination Compounds

1. A single word built from the **names of the ligands**, a prefix before each ligand to indicate **how many ligands** of that kind are present in the complex, and the **name of the central metal atom**.



hexa-cyano-ferrate(II)

2. The **positive ion** is named first, followed (after a space) by the name of the **negative ion**.

potassium hexa-cyano-ferrate(II)

3. Anionic ligands are named by replacing the usual ending with the suffix -o. The names of neutral ligands are unchanged.

CN⁻: cyanide

potassium hexa-cyano-ferrate(II)

4. Greek prefixes (di-, tri-, tetra-, penta-, hexa-) are used to specify the number of ligands of a given type attached to the central ion. If the name of the ligand itself contains a term such as mono- or di-, then the name of the ligand is placed in parentheses and the prefixes bis-, tris-, and tetrakis- are used instead of di-, tri-, and tetra-.

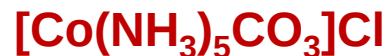
potassium hexa-cyano-ferrate(II)



Tris(ethylenediamine)platinum(IV) bromide

Naming Coordination Compounds

5. The **ligands are listed in alphabetical order**, without regard for the prefixes



Penta-amminecarbonatocobalt(III) chloride

6. A Roman numeral, enclosed in parentheses placed immediately **after the name of the metal, specifies the oxidation state**. If the complex ion has a net negative charge, the ending -ate is added to the stem of the name of the metal.



potassium hexa-cyano-ferrate(II)



Tris(ethylenediamine)platinum(IV) bromide

Naming Coordination Compounds

T A B L E 8.4

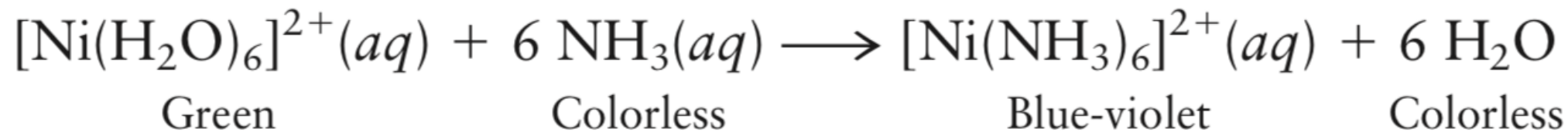
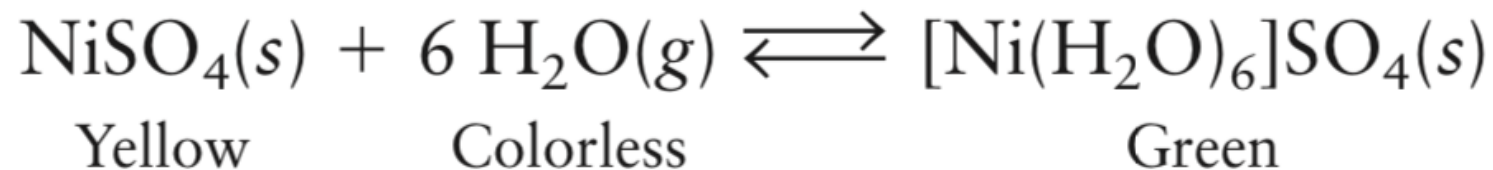
Examples of Complexes and Their Systematic Names

Complex	Systematic Name
$K_3[Fe(CN)_6]$	Potassium hexacyanoferrate(III)
$K_4[Fe(CN)_6]$	Potassium hexacyanoferrate(II)
$Fe(CO)_5$	Pentacarbonyliron(0)
$[Co(NH_3)_5CO_3]Cl$	Penta-amminecarbonatocobalt(III) chloride
$K_3[Co(NO_2)_6]$	Potassium hexanitrocobaltate(III)
$[Cr(OH_2)_4Cl_2]Cl$	Tetra-aquadichlorochromium(III) chloride
$[Pt(NH_2CH_2CH_2NH_2)_3]Br_4$	Tris(ethylenediamine)platinum(IV) bromide
$K_2[CuCl_4]$	Potassium tetrachlorocuprate(II)

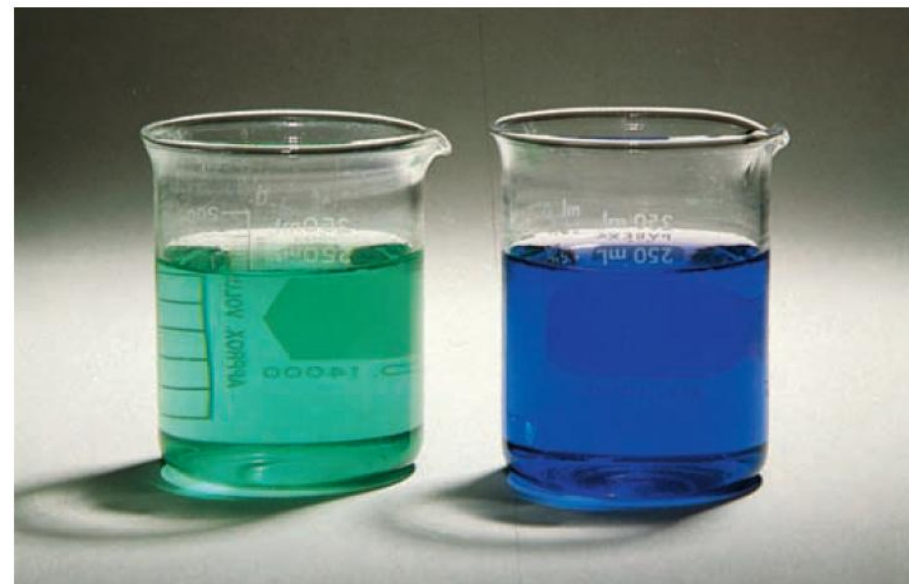
Interpret the names and write the formulas of these coordination compounds:

- (a) sodium tricarbonatocobaltate(III)
- (b) diamminediaquadichloroplatinum(IV) bromide
- (c) sodium tetranitratoborate(III)

Ligand Substitution (取代) Reactions



Labile (不稳定的) complexes: ligand substitution reactions proceed rapidly. Those in which substitution proceeds slowly or not at all are **inert** (惰性的). These terms are used to describe the different kinetics (rates) of the reactions.



Major points of today's lecture

- **Bonding in transition metal compounds and coordination complexes**
 1. **Chemistry of the Transition Metals**
 2. **Introduction to Coordination Chemistry**
 3. **Structures of Coordination Complexes**
 4. **Crystal Field Theory: Optical and Magnetic Properties**
 5. **Optical Properties and the Spectrochemical Series**
 6. **Bonding in Coordination Complexes**

Structures of Coordination Complexes: Octahedral Geometries

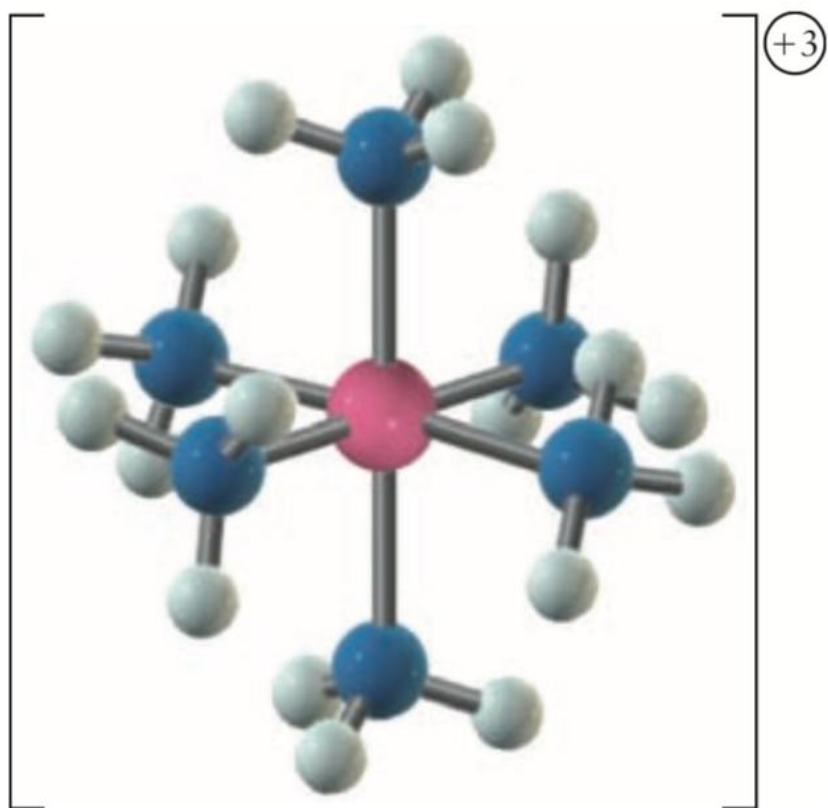
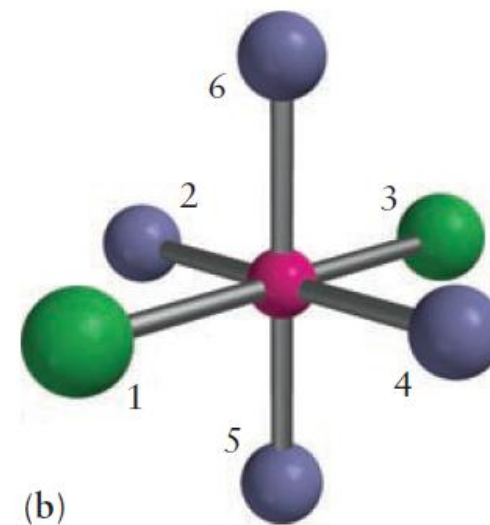
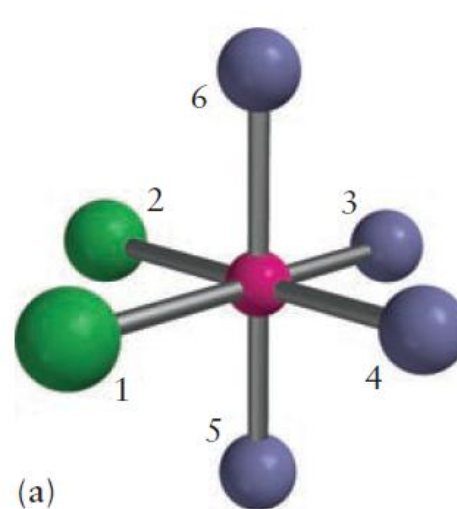


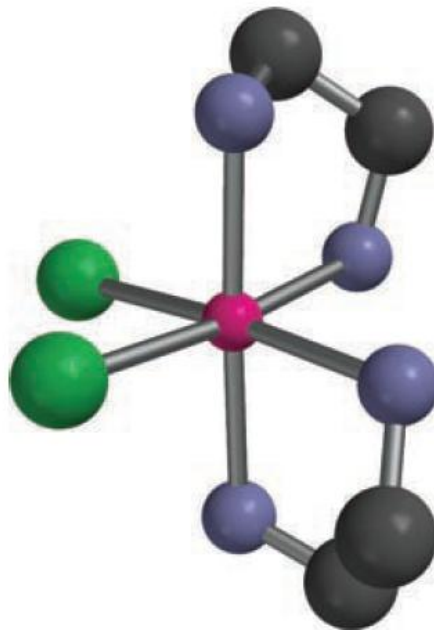
FIGURE 8.14 Octahedral structure of the $\text{Co}(\text{NH}_3)_6^{3+}$ ion. All six corners of the octahedron are equivalent.

$\text{cis-}[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ $\text{trans-}[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$

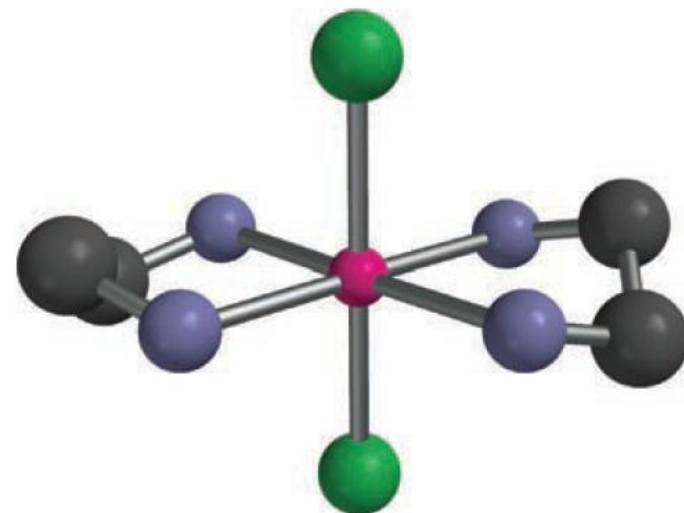


Structures of Coordination Complexes: Octahedral Geometries

FIGURE 8.10 The complex ion $[\text{CoCl}_2(\text{en})_2]^+$ is an octahedral complex that has *cis* and *trans* isomers, according to the relative positions of the two Cl^- ligands. Salts of the *cis* isomers are purple, and salts of the *trans* isomers are green.



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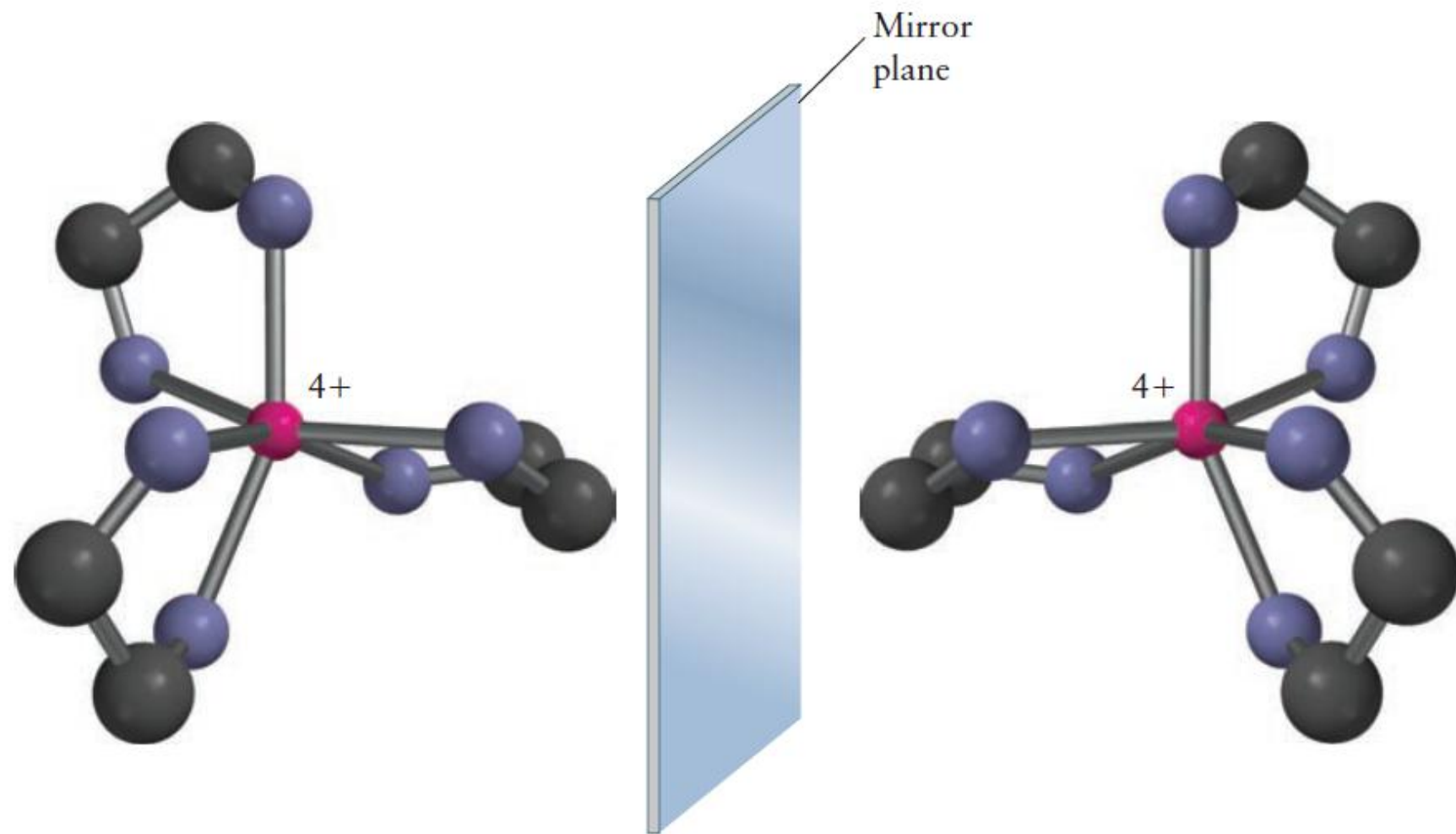


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How many geometric isomers does the octahedral coordination compound $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ have?

Structures of Coordination Complexes: Chiral Structures

FIGURE 8.13 Enantiomers of the $[\text{Pt}(\text{en})_3]^{4+}$ ion. Reflection through the mirror plane transforms one enantiomer into the other. The two cannot be superimposed by simple rotation.



Suppose that the complex ion $[\text{Co}(\text{NH}_3)_2(\text{OH}_2)_2\text{Cl}_2]^+$ is synthesized with the two ammine ligands *cis* to each other, the two aqua ligands *cis* to each other, and the two chloro ligands *cis* to each other (Fig. 8.14a). Is this complex optically active?

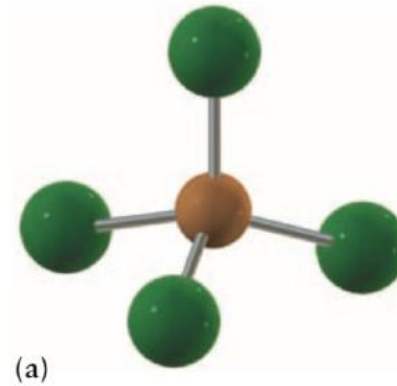


Solution

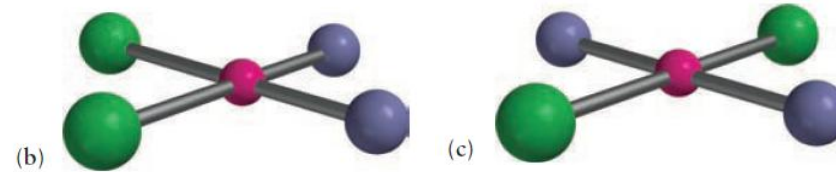
We represent a mirror by a shaded line and create the mirror image by making each point in the image lie at the same distance from the shaded line as the corresponding point in the original structure (see Fig. 8.14). Comparing the original (see Fig 8.14a) with the mirror image (see Fig. 8.14b) shows that *cis*, *cis*- $[\text{Co}(\text{NH}_3)_2(\text{OH}_2)_2\text{Cl}_2]^+$ is chiral, because the two structures cannot be superimposed even after they are turned. Although many chiral complexes contain chelating ligands, this example proves that nonchelates can be chiral.

Structures of Coordination Complexes: Square-Planar, Tetrahedral, and Linear

tetrahedral geometry: predominates in the early transition metals (those toward the left side of the d block of elements in the periodic table).



square-planar geometry: is common for four-coordinate complexes of Au^3 , Ir^+ , Rh^+ , and especially common for ions with the d^8 configurations (Ni^{2+} , Pd^{2+} , and Pt^{2+})



(b, c) Square planar, illustrating (b) the *cis* and (c) the *trans* forms of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$.

linear complexes: coordination numbers of 2, especially common for ions with d^{10} configurations (Cu^{2+} , Ag^+ , Au^+ , and Hg^{2+})

Homework: read “principle of modern chemistry”, chapter 8